

POOJA GARG

Full-Stack Engineer · Developer Tooling · AI & Data Pipelines · [LinkedIn](#) · [Portfolio](#)

Vogeliusweg 27c, 33100 Paderborn, Germany | poojagarg1903@gmail.com | +49 1778917535

PROFESSIONAL SUMMARY

Full-stack software engineer with 3+ years of experience in fintech and AI-driven developer tooling. Builds internal tools, React frontends, and Python/FastAPI services that reduce manual effort and accelerate engineering workflows. Proven impact: up to 80% reduction in manual configuration effort and 50% faster dataset inspection cycles. Currently pursuing an M.Sc. in Computer Science at the University of Paderborn, Germany.

PROFESSIONAL EXPERIENCE

Working Student in AI | dSPACE GmbH

Jun 2024 – Present · Paderborn, Germany

- Architected and owned the full React frontend for a production scenario configurator tool image-based coordinate marking, dynamic heuristic parameter handling, Plotly.js + Material UI connected via REST to a FastAPI backend and adopted across the data engineering team
- Integrated multimodal embedding models (multiple CLIP variants) into FiftyOne to enable similarity-based visualisation and efficient image analysis, with t-SNE projections for cluster inspection, reducing dataset inspection cycles by ~50%.
- Built a labelled dataset from dSPACE sensor car recordings using FiftyOne, and implemented additional enrichment features to attach metadata weather conditions, road conditions, and GNSS/IMU data to each entry.
- Containerised and deployed the full-stack system with Docker and Nginx, modularising components for a scalable, maintainable, and production-ready workflow reduced manual configuration effort by ~80%.
- Built automated Python pipelines for TFRecord dataset cards in TensorFlow including data lineage tracking, split statistics, and class distribution computation with HuggingFace-compatible YAML metadata output and a pytest suite for CI robustness.
- Wrote and maintained pytest test suites across multiple internal dSPACE repositories, achieving 80%+ code coverage across Python pipeline and API modules integrated into Azure DevOps for automated regression testing.
- Deployed versioned HTML and API documentation for an internal dSPACE tool using MkDocs and mike.

Software Engineer | PayU (Fintech)

Mar 2022 – Sep 2023 · Pune, India

- Developed and maintained PHP (CodeIgniter) payment services with MySQL-backed transaction processing for enterprise clients including automotive and large consumer platforms.
- Designed transaction workflows, payment wrappers, and service layers ensuring auditability, traceability, and compliance with financial requirements.
- Built RESTful APIs in CodeIgniter for internal systems and secure third-party integrations, with full request validation and structured error handling.
- Developed React frontends (React Router, Redux, Axios, Jest) for payment and admin portals, reducing manual operational effort by ~65%.
- Automated data retrieval and secure SFTP transfer workflows in Python, reducing manual reporting overhead by ~60%.

Software Engineer Intern | PayU (Fintech)

Sep 2021 – Mar 2022 · Pune, India

- Contributed to payment processing systems using Python, PHP, and MySQL, supporting daily transaction workflows.
- Assisted with API integrations and frontend/backend components, helping reduce development turnaround time by ~25%.
- Worked in an Agile, Git-based environment with cross-functional teams on payment and authentication APIs.

EDUCATION

M.Sc. Computer Science

· University of Paderborn

Oct 2023 – Present · Paderborn, Germany

B.Sc. Information Technology

· Savitribai Phule Pune University

Sep 2018 – Oct 2022 · Pune, India

Grade: 1.4 (German scale) or 9.06 CGPA

TECHNICAL SKILLS

Frontend: React, Redux, JavaScript (ES6+), TypeScript, HTML5, Tailwind CSS, Material UI, Plotly.js, Jest, Vite, npm/yarn

Backend: Python, FastAPI, PHP (CodeIgniter), Node.js, REST API design

Data & AI: FiftyOne, TFRecord, t-SNE, CLOOB embeddings, LiDAR/camera/GPS-IMU pipelines, multimodal data processing

Databases: MySQL

DevOps: Docker, Nginx, Git, GitLab, Bitbucket, Linux, MkDocs, CI/CD

Design: Figma, Design Systems, UX/UI collaboration

Methods: Agile / Scrum, Jira, Confluence

LANGUAGES

English — Proficient (C1) **German** — Elementary (A2) **Hindi** — Native

RESEARCH & CERTIFICATIONS

- Pawar, P. Y., Shaikh, F., Garg, P., Rokde, K., & Shelke, O. (2022). Detection of Phishing Website Using Machine Learning. *International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)*, 2(3), 766–770. <https://doi.org/10.48175/IJARSCT-3850>
- Certified: Capstone — Retrieving, Processing & Visualising Data with Python (Coursera / University of Michigan).[\[LINK\]](#)
- React — Advanced Development (Udemy).[\[LINK\]](#)